

# Supplementary Material: Lite Learning: Efficient Crop Classification in Tanzania Using Feature Extraction with Machine Learning & Crowd Sourcing

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## S-I. TIME SERIES FEATURE DEFINITIONS

This document is the online supplement to the main article “*Lite Learning: Efficient Crop Classification in Tanzania Using Feature Extraction with Machine Learning & Crowd Sourcing.*” It provides the full list of time-series features referenced in the main text (Methods, “Time Series Features”).

The features were extracted on a per-pixel basis from the 2023 Sentinel-2 monthly composites (bands B2, B6, B11, B12, the Enhanced Vegetation Index (EVI), and hue) using the **xr\_fresh** toolkit (Mann, Michael L. 2024). They summarize the temporal dynamics of crop growth and development across the growing season, spanning basic statistical descriptors, changes over time, and distribution-based metrics. Table S1 lists each statistic, a short description, and its defining equation. Throughout,  $x_i$  denotes the value of the band or index at time step  $i$  in a pixel’s monthly time series of length  $n$ ,  $\bar{x}$  its mean,  $\sigma$  its standard deviation,  $\mu_4$  its fourth central moment, and  $l$  an autocorrelation lag.

TABLE S1: Time-series features extracted with **xr\_fresh**, with descriptions and defining equations.

| Statistic                            | Description                                                      | Equation                                                                      |
|--------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Absolute energy                      | sum over the squared values                                      | $E = \sum_{i=1}^n x_i^2$                                                      |
| Absolute Sum of Changes              | sum over the absolute value of consecutive changes in the series | $\sum_{i=1}^{n-1}  x_{i+1} - x_i $                                            |
| Autocorrelation (1, 2 & 3 month lag) | Correlation between the time series and its lagged values        | $\frac{1}{(n-l)\sigma^2} \sum_{t=1}^{n-l} (x_t - \bar{x})(x_{t+l} - \bar{x})$ |
| Count Above Mean                     | Number of values above the mean                                  | $N_{\text{above}} = \sum_{i=1}^n (x_i > \bar{x})$                             |
| Count Below Mean                     | Number of values below the mean                                  | $N_{\text{below}} = \sum_{i=1}^n (x_i < \bar{x})$                             |
| Day of Year of Maximum Value         | Day of the year when the maximum value occurs in series          | —                                                                             |
| Day of Year of Minimum Value         | Day of the year when the minimum value occurs in series          | —                                                                             |
| Kurtosis                             | Measure of the tailedness of the time series distribution        | $G_2 = \frac{\mu_4}{\sigma^4} - 3$                                            |

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TABLE S1: Time-series features extracted with `xr_fresh`, with descriptions and defining equations.

| Statistic                               | Description                                                            | Equation                                                                                 |
|-----------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Linear Time Trend                       | Linear trend coefficient estimated over the entire time series         | $b = \frac{\sum_{i=1}^n (t_i - \bar{t})(x_i - \bar{x})}{\sum_{i=1}^n (t_i - \bar{t})^2}$ |
| Longest Strike Above Mean               | Longest consecutive sequence of values above the mean                  | —                                                                                        |
| Longest Strike Below Mean               | Longest consecutive sequence of values below the mean                  | —                                                                                        |
| Maximum                                 | Maximum value of the time series                                       | $x_{\max}$                                                                               |
| Mean                                    | Mean value of the time series                                          | $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$                                                 |
| Mean Absolute Change                    | Mean of absolute differences between consecutive values                | $\frac{1}{n-1} \sum_{i=1}^{n-1}  x_{i+1} - x_i $                                         |
| Mean Change                             | Mean of the differences between consecutive values                     | $\frac{1}{n-1} \sum_{i=1}^{n-1} (x_{i+1} - x_i)$                                         |
| Mean Second Derivative Central          | measure of acceleration of changes in a time series data               | $\frac{1}{2(n-2)} \sum_{i=1}^{n-2} \frac{1}{2} (x_{i+2} - 2 \cdot x_{i+1} + x_i)$        |
| Median                                  | Median value of the time series                                        | $\tilde{x}$                                                                              |
| Minimum                                 | Minimum value of the time series                                       | $x_{\min}$                                                                               |
| Quantile (q = 0.05, 0.95)               | Values representing the specified quantiles (5th and 95th percentiles) | $Q_{0.05}, Q_{0.95}$                                                                     |
| Ratio Beyond r Sigma (r=1,2,3)          | Proportion of values beyond r standard deviations from the mean        | $P_r = \frac{1}{n} \sum_{i=1}^n ( x_i - \bar{x}  > r\sigma)$                             |
| Skewness                                | Measure of the asymmetry of the time series distribution               | $\frac{n}{(n-1)(n-2)} \sum \left( \frac{x_i - \bar{x}}{\sigma} \right)^3$                |
| Standard Deviation                      | Standard deviation of the time series                                  | $\sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$                                      |
| Sum Values                              | Sum of all values in the time series                                   | $S = \sum_{i=1}^n x_i$                                                                   |
| Symmetry Looking                        | Measures the similarity of the time series when flipped horizontally   | $ \bar{x} - \tilde{x}  < r \cdot (x_{\max} - x_{\min})$                                  |
| Time Series Complexity (CID CE)         | measure of number of peaks and valleys                                 | $\sqrt{\sum_{i=1}^{n-1} (x_{i+1} - x_i)^2}$                                              |
| Variance                                | Variance of the time series                                            | $\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$                                  |
| Variance Larger than Standard Deviation | check if variance is larger than standard deviation                    | $\sigma^2 > 1$                                                                           |

## S-II. COMPUTATIONAL COMPLEXITY OF DEEP-LEARNING BASELINES

This section expands the efficiency comparison in the main text (Methods, “Computational Efficiency”), where the LightGBM “lite learning” model trains in 10–15 minutes on a CPU-only laptop. By contrast, the deep-learning architectures commonly applied to satellite image time series carry substantially higher training-time and memory costs. As in the main text,  $n$  denotes the number of training samples.

Recurrent models such as LSTMs, frequently used for temporal land cover analysis, exhibit complexity:

$$T_{LSTM} = O(e \cdot n \cdot t \cdot (h^2 + h \cdot i))$$

where  $e$  is the number of epochs,  $t$  the sequence length,  $h$  the hidden dimensionality, and  $i$  the input feature dimension. The quadratic dependence on  $h$  and the inherently sequential nature of LSTMs prevent efficient parallelization across the temporal axis, often extending training to several hours.

Spatial architectures further escalate these demands. A 2D U-Net scales linearly with the number of input channels (treating time steps as channels), but a 3D U-Net, which treats the temporal dimension as a third spatial axis, follows:

$$T_{3DUNet} = O\left(e \cdot n \cdot \sum_l k_l^3 \cdot c_{l-1} \cdot c_l \cdot s_l\right)$$

where the sum runs over network layers  $l$  with kernel size  $k_l$ , input and output channel counts  $c_{l-1}$  and  $c_l$ , and spatial size  $s_l$ . The cubic kernel term ( $k_l^3$ ) means a  $3 \times 3 \times 3$  convolution requires 27 operations per voxel, compared to 9 for a 2D equivalent. Furthermore, the memory (VRAM) required to store 4D tensors (Space  $\times$  Time  $\times$  Channels) for large-scale mapping routinely necessitates high-end GPU clusters.

## REFERENCES

Mann, Michael L. 2024. “xr\_fresh: Python Package for Feature Extraction from Raster Data Time Series.” <https://doi.org/10.5281/zenodo.12701466>.